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# Nonlinear absorption dynamics simulated in internal modification of glass at 532nm and 1064nm by ultrashort laser pulses

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**Abstract**

Dynamic ionization process at high pulse repetition rate is analysed based on the model consisting of rate equation for free-electrons, Gaussian beam propagation equation, thermal conduction and thermal ionization, showing the plasma generated near the geometrical focus moves toward the laser source periodically by the contribution of avalanche ionization. The effects of laser wavelength on the ionization process are discussed by comparing the contributions of multiphoton ionization and avalanche ionization. The crack tendency of 532nm and 1064nm is also compared between single pulse and multipulse irradiation.



*Keywords:* internal modificatin; thermal ionization; nonlinear ionization process; thermal conduction model; ultrashort laser pulses; glass; crack

## 1. Introduction

Internal modification of transparent dielectrics by ultrashort laser pulses (USLPs) has attracted much attention through applications including cell surgery [1], microwelding [2], waveguide formation [3] and selective etching [4]. Despite many papers published for the ionization process with the single-pulse irradiation [5,6], the ionization process with multipulses at high pulse repetition rates is limited because the ionization process is complicated by heat accumulation.

The heat accumulation effect in internal modification with USLPs was first reported by Schaffer et al [7] using purely thermal diffusion model; they showed that the multipulse-modified region can be much larger than single-pulse when the pulse repetition rate f is fast enough to restrict the heat diffusion from the focus volume. It is reported, however, that the nonlinear absorptivity increases with f by experiment and simulation [8], suggesting the ionization process is different from the single pulse. Recently, plasma behavior was analyzed by video camera [9,10] and simulation [9]; small plasma(s) moves dynamically upward to cover the region much larger than the focus volume, suggesting the importance of thermal ionization [1,11,12,13]. Despite some preliminary study [9], the simulation of the ionization process in heat accumulation regime is challenging, because the ionization process is affected by the thermal ionization at larger f.

In this study, dynamic ionization process is simulated based on the rate equation model for free-electrons coupled with the thermal conduction model, and thereby the effects of laser wavelength on the ionization process is discussed.

## 2. Simulation model

The evolution of free-electron density $\rho(r,z,t)$ under the influence of USLPs is given by the rate equation [1,12,13],


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$$\frac{\partial \rho(r,z,t)}{\partial t} = \eta_{photo} I^K(r,z,t) + \eta_{casc}\rho(r,z,t) I(r,z,t) - \eta_{casc}\rho(r,z,t)^2 \quad (1)$$

where $\eta_{photo}$ = photoionization coefficient, $\eta_{casc}$ = cascade ionization coefficient, $\eta_{rec}$ = recombination rate, K = number of photon for photoionization, I(r,z,t) = laser intensity in plasma. Cascade ionization coefficient $\eta_{casc}$ is given by [1,13]

$$\eta_{casc} = \frac{\tau}{\omega_L^2 \tau^2 + 1} \cdot \frac{e^2}{m c n_0 \varepsilon_0 (3/2) E_g} \quad (2)$$

where c = light speed in vacuum, e = electron charge, $\varepsilon_0$ = vacuum permittivity, m = electron mass, $n_0$ = refractive index of the glass at frequency $\omega_L$, $\tau$ = time between electron collision and $E_g$ = band gap energy.

Figure 1 shows the photoionization rate $\eta_{photo}$ at 532 nm and 1064 nm for borosilicate glass D263 ($E_g$ = 3.7 eV [8]) plotted vs. Keldysh paramter $\gamma$ [14]. Within our study conditions ($\gamma > 2$), multiphoton ionizaton (MPI) is dominant, and $\eta_{mpi}$ is approximated by a linear function of $\gamma$ [12] shown by a dashed line in Fig. 1 where $\eta_{mpi}$ at 532 nm (K = 2) is several orders larger than 1064 nm (K = 4). On the other hand, the avalanche ionization rate at 532 nm by Eq.(2) is about four times smaller than 1064 nm, suggesting MPI and cascade ionizations are dominant at 532 nm and 1064 nm, respectively.

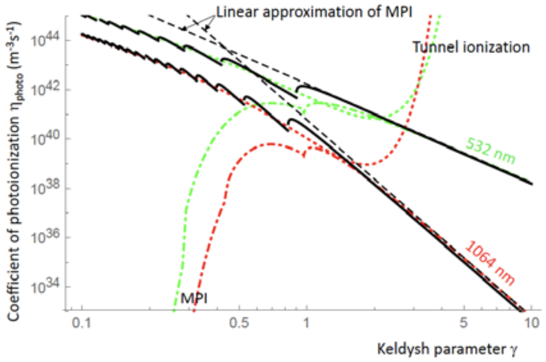


Fig. 1. Photoionization coefficient of borosilicate glass D263 having $E_g$ = 3.7 eV [8] calculated by Keldysh model [14] at 532 nm and 1064 nm. Multiphoton ionization (MPI) can be approximated by a dashed line.

Assuming the Gaussian beam [15] is fosuced into the bulk glass with the absorption coefficient α(r,z,t) at z = $z_f$, the laser intensity I(r,z,t) is given by

$$I(r,z,t) = \frac{2Q_0}{\pi \omega(z)^2 \tau_p} \exp\left[-\frac{2r^2}{\omega(z)^2}\right] \exp\left[-4\ln 2\left\{\frac{t - n_0(z-z_f)/c - 2\tau_p}{\tau_p}\right\}^2\right] - \int_0^z \alpha(r,z,t)\,dz, \quad (3)$$

Here $\tau_p$ is pulse width, $Q_0$ is pulse energy, and α(r,z,t) is

$$\alpha(r,z,t) I(r,z,t) = \eta_{photo} I^K(r,z,t) E_g + \eta_{casc}\rho(r,z,t) I(r,z,t)(3/2)E_g \quad (4)$$

where ω(z) is $1/e^2$-radius of the incident laser beam spot at z in glass with α(r,z,t) = 0 given by [15]

$$\omega(z) = \omega_0 \sqrt{1 + (z - z_f)^2 / z_r^2}. \quad (5)$$

Here $z_R$ = Rayleigh length and $\omega_0$ = radius at laser beam spot at waist with α(r,z,t) = 0.

Assuming n pulses with the energy per unit z-length q(z,n) deposit instantaneously at a repetition rate of f, we have derived analytical solution of temperature distribution T(r,z,t,n) given by [9]

$$T(r,z,t,n) = \frac{1}{\pi c_g \rho_g} \sum_{i=1}^{n} \frac{1}{\sqrt{\pi k\{t-(i-1)/f\}}} \int_{-\infty}^{\infty} \frac{q(z',i)}{\Omega^2(z',i) + 8k\{t-(i-1)/f\}} \times \exp\left[-\frac{2r^2}{\Omega^2(z',i) + 8k\{t-(i-1)/f\}} - \frac{(z-z')^2}{4k\{t-(i-1)/f\}}\right] dz', \quad (6)$$

where Ω(z,n) is $1/e^2$-radius of the laser beam, and $c_g$, $\rho_g$ and k are specific heat, density and heat diffusivity of glass, respectively. We also assumed the thermal properties are independent of temperature, for simplicity, and neglected the thermal radiation loss, despite very high temperature locally reached at laser absorption region. Instead we used a rather larger effective thermal conductivity [17] of 2.5 W/Km in this study. While the temperature at the heat affected region is underestimated by introducing the larger effective thermal conductivity, the temperature at the interaction region is overestimated by the neglect of thermal radiation, suggesting the absolute values of the simulated temperatures are not important.

It should be noted that the radius of the laser absorption region Ω(z,n) is different from the incident laser beam ω(z) in plasma, while Ω(z,n) = ω(z) was assumed in the existing papers [9,11]. We determined Ω(z,n) in the plasma pulse by pulse based on the nonlinear ionization processes using Eqs.(3) - (5), as will be shown in Figs. 5 and 6.

## 3. Experimental results

We used laser pulses having duration of 10 ps (FWHM) at wavelengths λ = 532 nm and 1064 nm. The laser beam was focused into the bulk glass at a depth of $z_f$ = 200 µm by an aspheric lens with a Galileo-telescope-like lens setup designed for the both wavelengths with NA of 0.5. The nonlinear absorptivity A is determined experimentally by [8]

$$A = 1 - Q_t / \left[Q_0 (1-R)^2\right] \quad (7)$$

where R is Fresnel reflectivity, $Q_0$ is incident pulse energy, $Q_t$ is transmitted pulse energy. The laser energy absorbed in the sample $Q_{ab}$ due to nonlinear process is simulated by

$$Q_{ab}(z) = Q_0 - 2\pi \int_0^{\infty}\int_{-\infty}^{\infty} I(r,z,t)\, r\, dr\, dt. \quad (8)$$

The experimental and the simulated values of absorbed laser energy $Q_{ab}(z)$ are plotted vs. $Q_0$ in Fig. 2. The simulated values satisfactorily agree with the experimental values, when using the rather larger laser spot radii $\omega_0$ of 1.9 µm and 3.8 µm for 532 nm and 1064 nm, suggesting the alignment of the aspheric lens is not perfect. 532 nm provides significantly larger nonlinear absorptivity with smaller threshold energy for the nonlinear absorption than 1064 nm. The side views of the laser-irradiated region are also compared in the figure at same values of $Q_{ab}(z)$. Crack tendency with single pulse in terms of size and number is larger at 1064 nm than 532 nm. Interestingly, however, in multipulse irradiation, the crack tendency becomes oppositely higher at 532 nm [16].

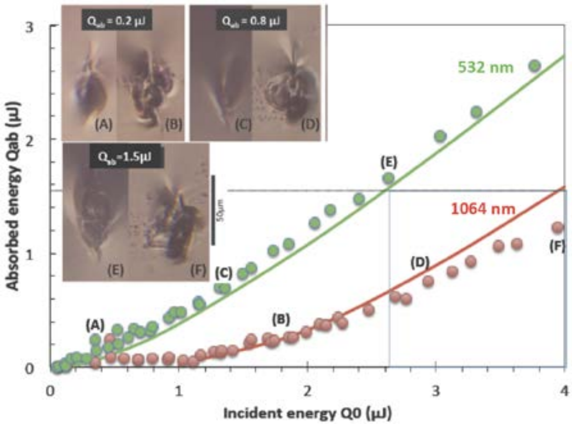


Fig. 2. Experimental and simulated values of $Q_{ab}(z)$ plotted vs. $Q_0$ with single pulse, assuming $\omega_0$ = 1.9 µm (532 nm) and $\omega_0$ = 3.8 µm (1064 nm) is focused into bulk glass. Side views of laser-irradiated region are also shown.

In the following sections, the ionization process is simulated at the same absorbed pulse energy of $Q_{ab}(z)$ = 1.55 µJ, which corresponds to the incident pulse energy of $Q_0$ = 2.6 µJ at 532 nm and $Q_0$ = 4 µJ at 1064 nm.

## 4. Ionization process in single and multipulses

### *4.1. Single pulse*

The simulated pulse energies of the transmission $q_t(z)$ and the absorption $q_{ab}(z)$ per unit z-length are plotted vs. z in Fig. 3. At 532 nm, $q_{ab}(z)$ spreads widely upstream up to z ≈ 100 µm due to larger $\eta_{mpi}$. At 1064 nm, on the other hand, $q_{ab}(z)$ concentrates to a narrow range of ≈ 20 µm, providing larger peak of $q_{ab}(z)$ and hence larger heat shock than 532 nm. Higher crack tendency at 1064 nm than 532 nm in single pulse observed in Fig. 2 is attributed to this heat shock difference.

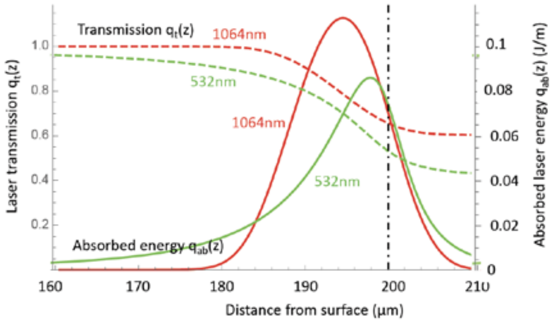


Fig. 3. Distributions of $q_t(z)$ and $q_{ab}(z)$ at 532 nm ($Q_0$ = 2.6 µJ) and 1064 nm ($Q_0$ = 4µJ) corresponding to $Q_{ab}$ = 1.55µJ at single pulse irradiation.

The ratio of multiphoton ionization to the total absorbed pulse energy $Q_{ab}$ can be estimated by

$$R_{mpi} = \frac{2\pi\eta_{mpi}E_g}{Q_{ab}} \int_0^\infty \int_0^\infty \int_{-\infty}^\infty I^K(r,z,t)\, r\, dr\, dz\, dt. \qquad (9)$$

$R_{mpi}$ increases rapidly as λ decreases (not show here), reaching 60 % at 532 nm. In contrast, $R_{mpi}$ is as small as ≈ 0.1 % at 1064 nm, indicating multiphoton ionization and avalanche ionization are dominant at 532 nm and 1064 nm, respectively.

### *4.2. Multipulses at high pulse repetition rate*

Figure 4 shows the simulated laser energy absorption $Q_{ab}$ plotted vs. the number of laser pulse n at f = 1MHz. The incident pulse energies are $Q_0$ = 2.6 µJ at 532 nm, and $Q_0$ = 4 µJ at 1064 nm corresponding to $Q_{ab}$ = 1.55 µJ. At 532 nm the laser energy is absorbed constantly with little variation. At 1064 nm, however, $Q_{ab}$ changes periodically at a cycle of $f_{cycle}$ = 130 ~ 140 pulses with increasing absorption peak with $f_{cycle}$, despite that the incident pulse energy $Q_0$ is constant.

Figure 5 shows the radial distribution of the specific laser absorption p(r,z,n), (laser energy absorbed per unit volume) at f = 1 MHz at different n, which is relate to $q_{ab}(z,n)$ by

$$q_{ab}(z,n) = 2\pi \int_0^\infty p(r,z,n)\, r\, dr. \qquad (10)$$

We approximated the radial distribution p(r,z,n) by a Gaussian curve (dashed curve) having $1/e^2$-radius of Ω(z,n) for Eq.(6). The approximation at 532 nm is nearly perfect with negligible effect of n, since the ionization at 532 nm is less affected by the free-electron density. At 1064 nm, however, the deviation of radial distribution of p(r,z,n) from the Gaussian curve increases, producing a dent near r = 0 due to the avalanche-dominated ionization, which depends on the existing free-electron density and hence the thermal ionization.

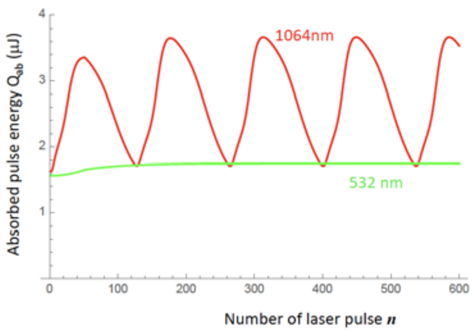


Fig. 4 Variation of $Q_{ab}$ plotted vs. number of laser pulse at 532 nm ($Q_0$ = 2.6 μJ) and 1064 nm ($Q_0$ = 4 μJ, f = 1MHz, D263).

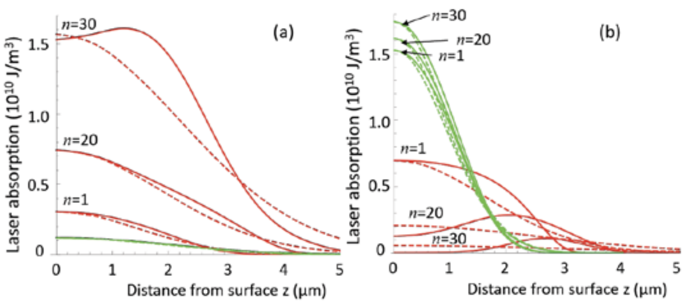


Fig. 5. p(r,z,n) vs. z at n = 1, 20 and 30 for 532 nm (green) and 1064 nm (red) by multipulse-irradiation at f = 1 MHz at (a) z = 186 μmand (b) z = 196 μm. Dashed shows their fitting to Gaussian curve.

In Fig. 6, the radius of absorbed region Ω(z,n) is plotted vs. z along with ω(z) for comparison. At 532 nm, Ω(z,n) shows the similar tendency as ω(z) (but somewhat smaller). At 1064 nm, however, Ω(z,n) is significantly smaller than ω(z) at n = 1. At n > 1, Ω(z,n) changes dynamically with n; the absorption front (shown by arrows; see Fig. 7) moves toward the laser source [13] nearly along the curve of Ω(z,n) at n = 1.

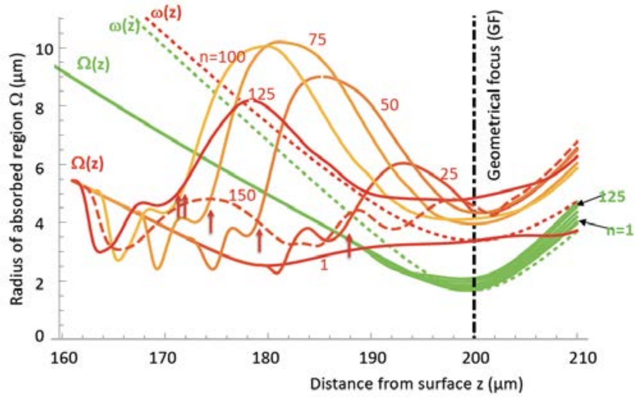


Fig. 6. Radii of laser absorption Ω(z,n) and incident laser beam ω(z) plotted vs. z at different number of laser pulse n for 532 nm and 1064 nm. The arrows indicate the locations of the absorption peak in Fig. 7(a).

Figure 7 shows the distribution of p(0,z,n) and temperature T(0,z,n) plotted along z-axis at different n just before the pulse impingement. At 532 nm, the peak is a little ahead of z = $z_f$ with small variation of the peak and locations. This is because the laser absorption is dominated by multiphoton ionization. At 1064 nm, they move forward with increasing their peaks at n = 25 – 50. $p_{peak}$ (the peak of p(0,z,n)) increases with n because the thermal ionization is strengthened by the heat accumulation. The peak location moves toward the laser source, because the avalanche ionization concentrates to the plasma front and shields downstream [1]. However, after reaching the highest value of $p_{peak}$ at n = 50 (z ≈ 180 μm), the heat accumulation increment is compensated by the decrease in the laser intensity, as the interaction point is apart from $z_f$.

When $p_{peak}$ decreases down to some critical value at n = 125 ($p_{peak}$ ≈ 4×$10^9$ W/m$^3$), the forward motion is finished. In this situation, the plasma is not hot enough any longer to provide sufficient electrons by thermal ionization, becoming transparent, so that a new plasma is produced near $z_f$ by the transmitted laser beam. It is noted two plasmas are observed simultaneously at n = 150, since the upstream plasma is still hot enough (but not hot enough for thermal ionization), in accordance with the experimental observation [9]. Thus the model explains qualitatively the periodic plasma motion at 1064 nm, as is seen in Fig. 7(b).

The laser pulse deposition can be observed experimentally using high-speed video camera as the shockwave [18], whose origin distributes in the region ahead of the geometrical focus where the laser deposition p(r,z,n) concentrates to a narrower region, as seen in Fig. 7(a).

Crack tendency is higher at 532 nm at high pulse repetition rates unlike the case of the single pulse [16], and is attributed to larger deposition of p(0,z,n) in narrower region repeatedly.

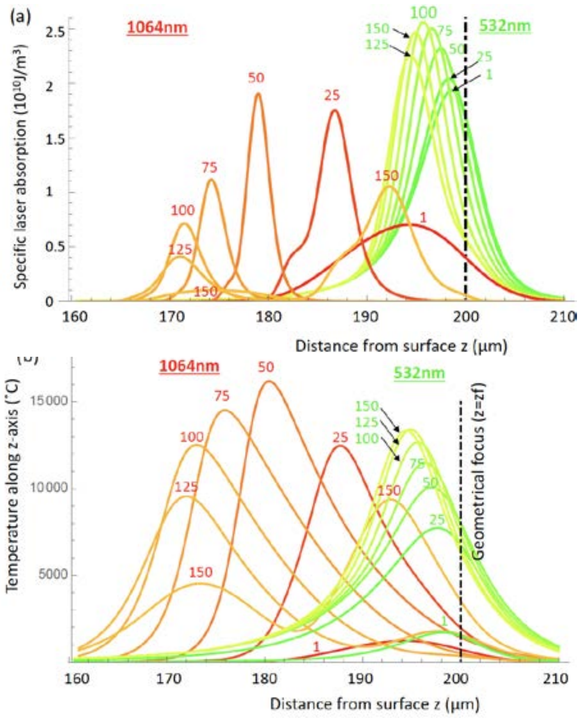


Fig. 7. Distribution of (a) p(0,z,t,n) and (b) T(0,z,t,n) at 1064 nm and 532 nm. ($Q_{ab}$ = 1.55μJ, f = 1MHz, D263). The laser energy deposition can be recognized by high-speed video camera as the origin of the shockwave [18].

## 5. Conclusions

Dynamic plasma motion in heat accumulation regime is analyzed at 532 nm and 1064 nm where multiphoton and avalanche ionizations are dominant, respectively. The plasma generated near the geometrical focus moves toward the laser source periodically by the contribution of avalanche ionization strengthened by thermal ionization at 1064 nm, while the plasma at 532 nm is rather static.

## Acknowledgements

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## References

[1] Vogel A, Novak J, Hüttman G, and Paltauf G, Mechanisms of femtosecond laser nanosurgery of cells and tissues, Appl. Phys. B **81**(8), 1015–1047 (2005).

[2] Miyamoto I, Cvecek K, Schmidt M. "Characteristics of laser absorption and welding in FOTURAN glass by ultrashort laser pulses", Optics Express, **19**, 22961-22973 (2011).

[3] Shah L, Arai AY, Eaton SM, Herman PR, Waveguide writing in fused silica with a femtosecond fiber laser at 522 nm and 1 MHz repetition rate, Opt. Express, **13**, 1999-2006 (2005).

[4] Wortmann D, Gottmann J, Brandt N, Horn-Solle H, Micro- and nanostructures inside sapphire by fs-laser irradiation and selective etching, Opt. Express, **16** (2008) 1517-1522.

[5] Nahen K, Vogel A, Plasma Formation in Water by Picosecond and Nanosecond Nd:YAG Laser Pulses—Part II: Transmission, Scattering, and Reflection, IEEE J. Quant. Elect., **2** 861-871 (1996).

[6] Arnold CL, Heisterkamp A, Ertmer W, and Lubatschowski H, Computational model for nonlinear plasma formation in high NA micromachining of transparent materials and biological cells, Opt. Express, **16**, 10303 (2007).

[7] Schaffer BC, Garcia JF, EMazur E, Bulk heating of transparent materials using high-repetition-rate femtosecond laser, Appl.. Phys. **A 76**, 187-208 (2003).

[8] Miyamoto I, Cvecek K, Schmidt M, Evaluation of nonlinear absorptivity in internal modification of bulk glass by ultrashort laser pulses, Opt. Express, **19**, 10714-10727 (2011).

[9] Miyamoto I, Okamoto Y, Tanabe R, Ito Y, Cvecek K, Schmidt. Mechanism of dynamic plasma motion in internal modification of glass by fs-laser pulses at high pulse repetition rate. Opt. Express 2016; 24:25718-31.

[10] Alexeev I, Heberle J, Cvecek K, Nagulin KY, Schmidt M, High Speed Pump-Probe Apparatus for Observation of Transitional Effects in Ultrafast Laser Micromachining Processes. Micromachines 2015;6:1914-22.

[11] Sun M, Eppelt U, Schulz W, Zhu J, Role of thermal ionization in internal modification of bulk borosilicate glass with picosecond laser pulses at high repetition rates, Opt. Mat. Express **3,** 1716 (2013).

[12] Kennedy PK, A first-order model for computation of laser-induced breakdown thresholds in ocular and aqueous media: Part I – Theory, IEEE J. Quant. Elect., **31** 2241-2249 (1995).

[13] Linz N, Freidank S, Liang X, Noak J, Paltauf G, Vogel A, "Roles of tunneling, multiphoton ionization, and cascade ionization for femtosecond optical breakdown in aqueous media," http://www.dtic.mil/dtic/tr/ fulltext/u2/a521817.pdf (2009.)

[14] Keldysh LV, Ionization in the field of a strong electromagnetic wave", Sov. Phys. JETP (1965) 1207-1314.

[15] Siegman AE, Lasers, University Science, Mill NVelly, Calif., (1986) Chap. 17.

[16] Cvecek K, Steglein F, Miyamoto I, Schmidt M, Influence of wavelength on glass weling by ultra-short laser pulses, J. Laser Micro/Nanoeng, **12** (2017) 115-119

[17] Kiyohashi H, Hayakawa N, Aratani S, Masuda H, Thermal codictovoty measurements of float glass at high temperatures by needle probe mechod, J. Cermaic Soc. of Japan (2000) 381-386

[18] Cvecek K, Miyamoto I, Heberle J, Bergler M, DeLigny D, Schmidt M, Analysis of shockwave formation in glass welding by ultrashort pulses. Proc. 10th Conf. on Photonic Tech – LANE 2018 (to be published).